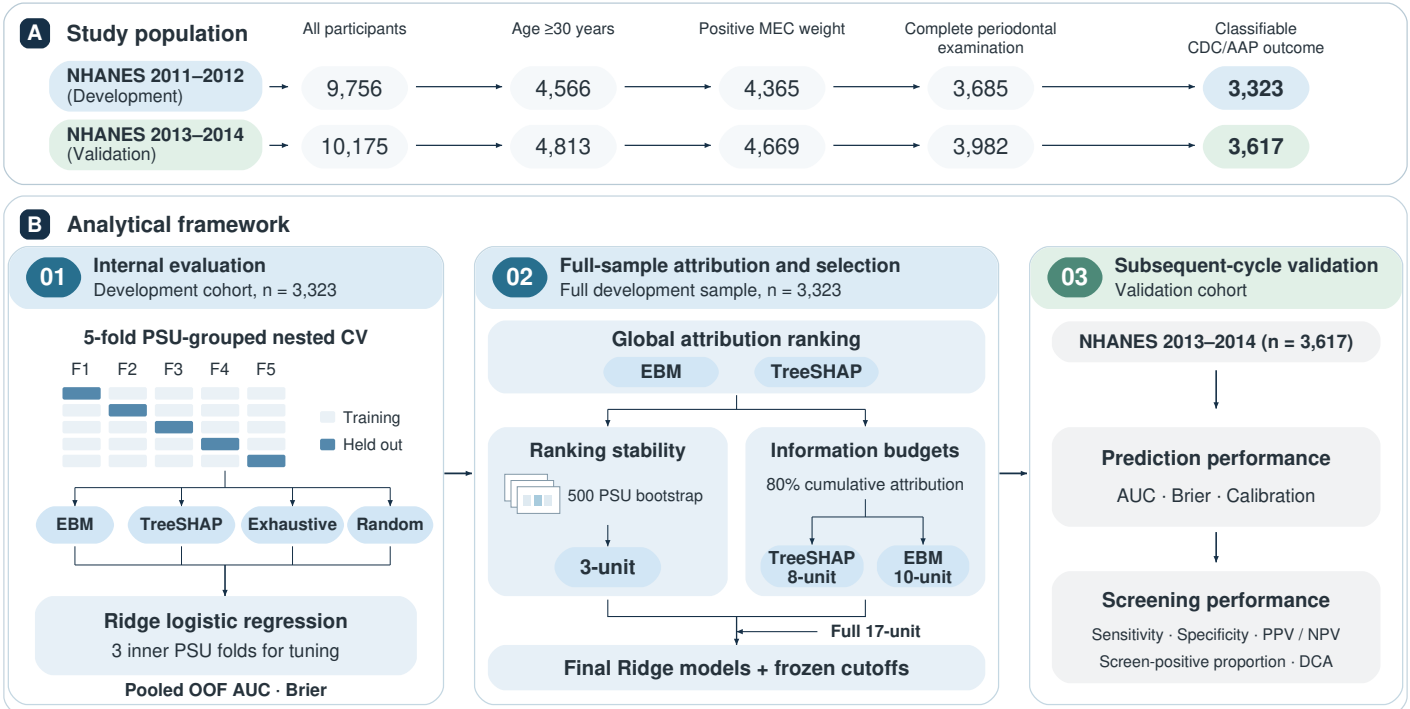

A
Study population
All participants
Age ≥30 years
Positive MEC weight
Complete periodontal examination
Classifiable CDC/AAP outcome
NHANES 2011–2012 (Development)
9,756
4,566
4,365
3,685
3,323
NHANES 2013–2014 (Validation)
10,175
4,813
4,669
3,982
3,617
B
Analytical framework
01
Internal evaluation
Development cohort, n = 3,323
5-fold PSU-grouped nested CV
F1
F2
F3
F4
F5
Training
Held out
EBM
TreeSHAP
Exhaustive
Random
Ridge logistic regression
3 inner PSU folds for tuning
Pooled OOF AUC · Brier
02
Full-sample attribution and selection
Full development sample, n = 3,323
Global attribution ranking
EBM
TreeSHAP
Ranking stability
500 PSU bootstrap
3-unit
Information budgets
80% cumulative attribution
TreeSHAP 8-unit
EBM 10-unit
Full 17-unit
Final Ridge models + frozen cutoffs
03
Subsequent-cycle validation
Validation cohort
NHANES 2013–2014 (n = 3,617)
Prediction performance
AUC · Brier · Calibration
Screening performance
Sensitivity · Specificity · PPV / NPV
Screen-positive proportion · DCA